

Lecture 11 — Compensated Coupling and Bayes Selection

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Meeting record. Lecture 11 (September 29, Zoom) introduced single-resource allocation and the hindsight comparison. Lecture 12 (October 1) was canceled. Lecture 13 (October 6) developed the uniform-value compensated-coupling proof. The finite-type Bayes Selector below is a *planned continuation for Lecture 14 (October 8)*. Spiral Down has separate notes.

11.1 Why compare online decisions with hindsight?

Lectures 9–10 gave an $O(\sqrt{T})$ additive guarantee for a static fluid policy. Their two-value example shows a matching gap between that particular policy and hindsight. Realized demand fluctuates around its mean; an online policy can respond to these fluctuations as they unfold. We ask:

When can an online decision agree with an optimal hindsight decision?

The argument here follows the compensated-coupling structure of Banerjee’s 2022 Simons Institute talk, specialized to the model taught in class.

11.2 One resource and three benchmarks

There are T periods and initial capacity $B \in \{0, \dots, T\}$. Buyer t has random value $V_t \geq 0$; the values are independent and identically distributed (iid) according to known F . After seeing V_t and the past, a policy chooses $A_t \in \{0, 1\}$, where 1 means accept. Each acceptance uses one unit. Its realized reward is $R^\pi = \sum_{t=1}^T A_t V_t$, with $\sum_t A_t \leq B$.

For realized values $v_{(1)} \geq \dots \geq v_{(T)}$ in descending order, the offline clairvoyant earns

$$\text{OPT}(v_1, \dots, v_T) = \sum_{i=1}^B v_{(i)}.$$

The empty sum is zero when $B = 0$; when $B = T$ it includes all buyers. The optimal online expected reward is $\sup_\pi \mathbb{E}[R^\pi]$. For a *particular* policy, expected regret means

$$\text{Reg}(\pi) = \mathbb{E}[\text{OPT}(V_1, \dots, V_T) - R^\pi]. \quad (11.1)$$

The clairvoyant, the best online policy, and a proposed online policy are distinct benchmarks.

Brief static-price review. For continuous F , a static threshold p_L satisfying $\mathbb{P}(V_t \geq p_L) = B/T$ gives $N(p_L) \sim \text{Binomial}(T, B/T)$ qualifying buyers. Its standard deviation is at most $\sqrt{T}/2$. For bounded values, this count error yields the $O(\sqrt{T})$ fluid comparison developed in Lectures 9–10; their two-value instance shows a matching gap for the specified static policy. The board shorthand $\mathbb{E}[\text{OPT}] \leq Bp_L$ is not generally true: hindsight may select buyers with values above p_L .

11.3 Compensated coupling: forcing hindsight

The Simons talk considers two controllers on one sample path. ONLINE knows current and past observations. OFFLINE knows the entire realized future. Whenever OFFLINE does not wish to take ONLINE's action, pay it the decrease in its optimal continuation value and force it to follow ONLINE.

Fix a complete sequence $\omega = (v_1, \dots, v_T)$. Let $H_t(b; \omega)$ be the sum of the $\min\{b, T - t + 1\}$ largest values from periods $t, \dots, T$, with $H_{T+1}(b; \omega) = 0$. Thus excess capacity simply remains unused. For feasible actions define

$$\begin{aligned} Q_t^{\text{off}}(b, 1; \omega) &= v_t + H_{t+1}(b - 1; \omega), \\ Q_t^{\text{off}}(b, 0; \omega) &= H_{t+1}(b; \omega), \quad H_t(b; \omega) = \max_{a \in \{0, 1\}: a \leq b} Q_t^{\text{off}}(b, a; \omega). \end{aligned} \quad (11.2)$$

This is $V^{\text{off}}(t, s)$ and its action-value $Q^{\text{off}}(t, s, a)$ from the talk, with state $s = (t, b)$. Both depend on the *realized* future; neither is the expected online dynamic-programming value.

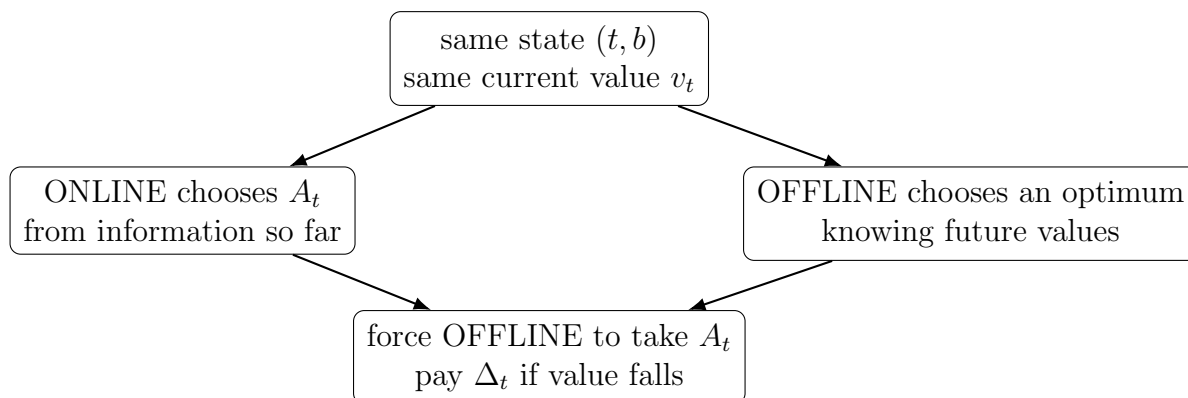


Figure 11.1. Compensated coupling evaluates ONLINE's action against the offline optimum at the *same* remaining capacity and realized future.

Let B_t be ONLINE's capacity before period t : $B_1 = B$, $B_{t+1} = B_t - A_t$. The smallest necessary compensation is

$$\Delta_t = H_t(B_t; \omega) - Q_t^{\text{off}}(B_t, A_t; \omega) \geq 0. \quad (11.3)$$

The chosen action earns $A_t v_t$ and reaches state $(t+1, B_{t+1})$, so

$$H_t(B_t; \omega) = A_t v_t + H_{t+1}(B_{t+1}; \omega) + \Delta_t.$$

Summing over t cancels successive continuation values. Since $H_1(B; \omega) = \text{OPT}(\omega)$ and $H_{T+1} = 0$,

$$\boxed{\text{OPT}(\omega) - R^\pi(\omega) = \sum_{t=1}^T \Delta_t(\omega)}, \quad \text{Reg}(\pi) = \sum_{t=1}^T \mathbb{E}[\Delta_t]. \quad (11.4)$$

The talk's more general comparison is an inequality for other benchmarks. With this exact hindsight optimum and minimal compensation, it is a *pathwise equality*. No formal information-flow notation is needed to verify the cancellation.

11.4 Satisfying actions and ties

An action is *satisfying* if some optimal hindsight allocation from the current state takes it. Equivalently, its Q_t^{off} value equals H_t , so $\Delta_t = 0$. Multiple actions can be satisfying. Predicting one arbitrarily tie-broken hindsight allocation would mistake a costless alternative for an error.

Example 11.1 (A marginal buyer). *There are two seats, the current value is 1, and future values are $(2, 1, 0)$. Hindsight earns 3 using either the current or the future value-1 buyer: accept and reject are both satisfying. With future $(2, 2, 0)$, only reject is satisfying; with future $(0, 0, 0)$, only accept is satisfying.*

If values are bounded by $v_{\max}$, then a non-satisfying action costs at most $v_{\max}$:

$$\text{Reg}(\pi) \leq v_{\max} \sum_{t=1}^T \mathbb{P}(A_t \text{ is not satisfying at ONLINE's state}). \quad (11.5)$$

For continuous values, measuring the *size* of the mismatch gives a sharper calculation.

11.5 Uniform values: the October 6 proof

Assume $V_t \stackrel{\text{iid}}{\sim} \text{Uniform}[0, 1]$. At the current period, let b seats remain and let $h = T - t$ be the number of buyers *strictly after* the current one. For $1 \leq b \leq h$, let $W_{h,b}$ be the b -th largest of those h future values. It is random to ONLINE but fixed on a realized future path. Equation (11.2) gives

$$Q_t^{\text{off}}(b, 1) - Q_t^{\text{off}}(b, 0) = V_t - W_{h,b}.$$

Thus hindsight accepts if $V_t > W_{h,b}$ and rejects if $V_t < W_{h,b}$. If $b = 0$, reject is forced; if $b \geq h + 1$, accept all remaining buyers. These boundary actions pay zero compensation.

Suppose ONLINE accepts when $V_t \geq \theta$, where θ depends only on (h, b) . A disagreement occurs only when V_t lies between the online threshold and the future order statistic. Its exact cost is

$$\Delta_t = |W_{h,b} - V_t| \mathbf{1}\{\min(\theta, W_{h,b}) < V_t < \max(\theta, W_{h,b})\}. \quad (11.6)$$

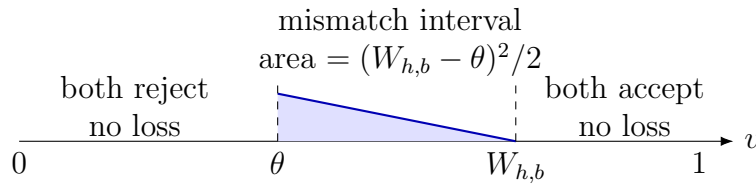


Figure 11.2. When $\theta < W_{h,b}$, ONLINE accepts a buyer whom hindsight rejects. The vertical height is the lost value $W_{h,b} - v$. Reversing the threshold order gives the same triangle.

Condition on $W_{h,b} = w$. The current uniform value is independent of the future. If $\theta < w$, integrate its density over the interval:

$$\mathbb{E}[\Delta_t \mid W_{h,b} = w] = \int_{\theta}^w (w - v) dv = \frac{(w - \theta)^2}{2}.$$

If $w < \theta$, the integral is $\int_w^{\theta} (v - w) dv = (\theta - w)^2/2$. Therefore

$$\boxed{\mathbb{E}[\Delta_t \mid W_{h,b}] = \frac{1}{2}(W_{h,b} - \theta)^2.} \quad (11.7)$$

The triangle's *area*, rather than the probability of disagreement alone, is the economic loss.

11.6 Order statistics and the logarithmic sum

The b -th largest among h uniforms is the $(h - b + 1)$ -st smallest, so $W_{h,b} \sim \text{Beta}(h - b + 1, b)$. Its moments are

$$\mathbb{E}[W_{h,b}] = \frac{h - b + 1}{h + 1} = 1 - \frac{b}{h + 1}, \quad \text{Var}(W_{h,b}) = \frac{b(h - b + 1)}{(h + 1)^2(h + 2)}. \quad (11.8)$$

For $b = 1$ this is the maximum, with mean $h/(h + 1)$; for $b = h$ it is the minimum, with mean $1/(h + 1)$.

For any deterministic θ chosen from state (h, b) ,

$$\mathbb{E}[(W_{h,b} - \theta)^2] = \text{Var}(W_{h,b}) + (\mathbb{E}[W_{h,b}] - \theta)^2.$$

Thus the exact minimizer of the one-step *expected compensation* is

$$\theta_{h,b} = \mathbb{E}[W_{h,b}] = 1 - \frac{b}{h+1}. \quad (11.9)$$

This is the uniform quantile at the remaining capacity fraction $b/(h+1)$ among the $h+1$ opportunities *including* the current buyer. The classroom shorthand $1 - b/h$ uses only the h future buyers; for $h \geq 1$ it differs by $b/[h(h+1)] \leq 1/(h+1)$. The exact choice makes the squared-error calculation clean.

Since $b(h-b+1) \leq (h+1)^2/4$, equations (11.7) and (11.8) give

$$\mathbb{E}[\Delta_t \mid B_t = b] = \frac{1}{2} \text{Var}(W_{h,b}) \leq \frac{1}{8(h+2)} \quad (1 \leq b \leq h). \quad (11.10)$$

The future values remain independent of B_t , which depends only on the past.

Theorem 11.2 (Uniform-value threshold policy). *Suppose values are iid uniform on $[0, 1]$, $0 \leq B \leq T$, and the policy rejects if $b = 0$, accepts if $b \geq h+1$, and otherwise accepts exactly when $V_t \geq 1 - b/(h+1)$. Then*

$$\text{Reg}(\pi) \leq \frac{1}{8} \sum_{h=1}^{T-1} \frac{1}{h+2} \leq \frac{1}{8} H_{T+1} = O(\log T), \quad (11.11)$$

where $H_n = \sum_{k=1}^n 1/k$.

Proof: Apply the pathwise identity (11.4), then (11.10) for each remaining horizon h . Boundary states have zero compensation. ■

This upper bound also applies to the best online policy. Bray's separate lower bound shows logarithmic regret is the right order for uniform values when capacity is a fixed nontrivial fraction of the horizon. This is a continuous-value result, not an $O(1)$ finite-type theorem.

Remark 11.3 (Mean versus median). *The rule (11.9) minimizes expected compensation. An ideal selector that minimizes only the probability of an unsatisfying action would compare V_t with a median of $W_{h,b}$. These thresholds need not coincide. The mean rule is the model-based quantile policy analyzed above.*

For general continuous F , the analogous capacity-fraction quantile is $F^{-1}(1 - b/(h+1))$ when defined. The future order statistic has distribution $F^{-1}(U_{(h-b+1)})$, whose mean need not equal this quantile. The uniform proof does not automatically establish $O(\log T)$ for every continuous F .

11.7 Planned October 8 continuation: finite types

This section was planned for Lecture 14 (October 8). The teaching record for September 29 and October 6 ends above.

Suppose values belong to a fixed set $0 < v_1 < \dots < v_K \leq v_{\max}$, with iid probabilities $p_j = \mathbb{P}(V_t = v_j) > 0$. Depending on future counts, the current value may be necessarily rejected, marginal, or necessarily accepted by hindsight. At a tie, both actions can be satisfying. This discrete model predicts a *set* of offline actions.

For current type j and future path ω , let $\mathcal{S}_{h,b,j}(\omega) \subseteq \{0, 1\}$ be the satisfying actions. The *ideal Bayes Selector* chooses a feasible action minimizing

$$q_{h,b,j}(a) = \mathbb{P}(a \notin \mathcal{S}_{h,b,j}(\omega) \mid B_t = b, V_t = v_j).$$

For two actions this resembles a majority rule, but it differs from predicting one tie-broken hindsight allocation.

A tractable finite-type version. Vera and Banerjee analyze the *Fluid Bayes Selector*. Relabel types in descending order $r_1 > \dots > r_K > 0$ with probabilities $p_1, \dots, p_K$, and put $\bar{p}_j = \sum_{i=1}^j p_i$, $\bar{p}_0 = 0$. Let $m = h + 1$ include the current opportunity. With $b > 0$ always accept the highest type. For an arrival of type $j > 1$, accept if

$$\frac{b}{m} \geq \bar{p}_{j-1} + \frac{p_j}{2}, \quad (11.12)$$

and reject otherwise; with $b = 0$, reject. This rule solves the remaining fluid LP and accepts type j when its predicted acceptance count is at least half of expected type- j demand. It minimizes suitable *upper bounds* on disagreement probabilities. The ideal selector minimizes the probabilities themselves.

Theorem 11.4 (Finite-type constant regret; Vera–Banerjee). *For one resource, K fixed distinct positive value types, iid multinomial arrivals with $p_j > 0$, and the Fluid Bayes Selector (11.12), expected regret against the hindsight optimum satisfies*

$$\text{Reg}(\pi_{\text{fluidBS}}) \leq 2v_{\max} \sum_{j=2}^K \frac{1}{p_j} \leq \frac{2(K-1)v_{\max}}{p_{\min}}, \quad p_{\min} = \min_j p_j. \quad (11.13)$$

This is independent of T and B . The source also obtains the bound for the exact-probability Bayes Selector.

Here is why the scale is constant. If (11.12) rejects type j when hindsight needs it, the realized count of types $1, \dots, j$ is unusually low. If it accepts when hindsight does not need it, the count of types $1, \dots, j-1$ is unusually high. The rule leaves a gap of $p_j/2$ between

the capacity fraction and the relevant expected count. A binomial tail bound makes either non-satisfying event at remaining horizon m at most $e^{-mp_j^2/2}$. Multiplying by the arrival probability p_j and summing over m produces a constant of order $1/p_j$. The full accounting gives (11.13); see Theorem 2 and Appendix B.3 of the source.

The probabilities stay fixed as T grows. If a type becomes rarer with T , the displayed constant may grow.

11.8 Takeaways

1. Compensation is the reduction in hindsight value from forcing an online action; the reductions telescope pathwise.
2. A satisfying action pays zero compensation, even if a different hindsight-optimal action also exists.
3. For uniform values, conditional expected compensation is the triangle area $(W_{h,b} - \theta)^2/2$.
4. The exact mean threshold $1 - b/(h+1)$ has $O(1/h)$ one-step loss and $O(\log T)$ cumulative regret.
5. For fixed finite types, the Fluid Bayes Selector has $O(1)$ regret under Theorem 11.4. The $O(\sqrt{T})$ static, $O(\log T)$ uniform, and $O(1)$ finite-type statements concern their specified policies and assumptions.

Recommended reading

Lecture sources.

Banerjee, *The Compensated Coupling*, IFML–Simons Institute talk (October 2022), slides 14–20 and 24–34; September 29 Zoom notes and October 6 board.

Finite types.

Vera and Banerjee, *The Bayesian Prophet: A Low-Regret Framework for Online Decision Making*, *Management Science* 67 (2021), Sections 3–4, Theorem 2 and Appendix B.3 ([preprint](#)).

Continuous values*.

Bray, *Logarithmic Regret in Multisecretary and Online Linear Programs with Continuous Valuations*, *Operations Research* (2024), Section 3.